$$\text{This implies, } \frac{f(x+\Delta x)-f(x)}{\Delta x} = \frac{\dfrac{v(x)\left[u(x+\Delta x)-u(x)\right]}{\Delta x} - \dfrac{u(x)\left[v(x+\Delta x-v(x)\right]}{\Delta x}}{v(x+\Delta x)v(x)}$$

$$\text{This implies, } \lim_{\Delta x\to 0}\frac{f(x+\Delta x)-f(x)}{\Delta x} = \frac{v(x)\lim\limits_{\Delta x\to 0}\dfrac{u(x+\Delta x)-u(x)}{\Delta x} - u(x)\lim\limits_{\Delta x\to 0}\dfrac{v(x+\Delta x)-v(x)}{\Delta x}}{v(x)\lim\limits_{\Delta x\to 0}v(x+\Delta x)}$$

$$= \frac{v(x)u'(x)-u(x)v'(x)}{v(x)v(x)} \qquad \left(\because \lim_{\Delta x\to 0} v(x+\Delta x) = v(x)\right)$$

$$\text{This implies, } f'(x) = \frac{v(x)u'(x)-u(x)v'(x)}{\left[v(x)\right]^2}.$$

$$\text{That is, } \frac{d}{dx}(f) = \frac{vu'-uv'}{v^2}$$

or $\boxed{\dfrac{d}{dx}\left(\dfrac{u}{v}\right) = \dfrac{v\dfrac{du}{dx}-u\dfrac{dv}{dx}}{v^2}}$.

Theorem 10.5 (Chain Rule / Composite Function Rule or Function of a Function Rule)

Let $y=f(u)$ be a function of u and in turn let $u=g(x)$ be a function of x so that $y = f\left(g(x)\right) = (fog)(x)$.

Then $\dfrac{d}{dx}(f(g(x)) = f'(g(x))g'(x)$.

Proof

In the above function $u = g(x)$ is known as the inner function and f is known as the outer function. Note that, ultimately, y is a function of x.

Now $\Delta u = g(x+\Delta x)-g(x)$

$$\text{Therefore, } \frac{\Delta y}{\Delta x} = \frac{\Delta y}{\Delta u}\times\frac{\Delta u}{\Delta x} = \frac{f(u+\Delta u)-f(u)}{\Delta u}\times\frac{g(x+\Delta x)-g(x)}{\Delta x}.$$

Note that $\Delta u \to 0$ as $\Delta x \to 0$

$$\text{Therefore, } \lim_{\Delta x\to 0}\frac{\Delta y}{\Delta x} = \lim_{\Delta x\to 0}\left(\frac{\Delta y}{\Delta u}\times\frac{\Delta y}{\Delta x}\right)$$

$$= \lim_{\Delta u\to 0}\left(\frac{\Delta y}{\Delta u}\right).\lim_{\Delta x\to 0}\left(\frac{\Delta y}{\Delta x}\right)$$

$$= \lim_{\Delta u\to 0}\frac{f(u+\Delta u)-f(u)}{\Delta u}\times\lim_{\Delta x\to 0}\frac{g(x+\Delta x)-g(x)}{\Delta x}$$

$$= f'(u)\times u'(x)$$

$$= f'\left(g(x)\right)g'(x) \quad \text{or} \quad \boxed{\frac{d}{dx}\left(f(g(x)\right) = f'\left(g(x)\right)g'(x)}.$$

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